

POLTAVA, Ukraine

WMO: 335060

Lat: 49.6094N

Lon: 34.5447E

Elev: 525

StdP: 14.42

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.7	3.6	-7.8	3.7	-0.3	-2.8	4.8	4.3	22.9	16.3	20.7	20.0	6.0	50	0.491

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.4	89.7	68.2	86.4	67.2	83.5	66.2	71.1	83.8	69.6	82.0	68.2	79.7	6.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.9	100.9	76.5	65.4	95.7	75.0	63.9	90.7	73.7	35.2	83.9	33.9	82.1	32.7	79.7	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.8	16.9	14.4	-7.3	93.8	6.6	3.6	-12.0	96.4	-15.9	98.5	-19.5	100.5	-24.3	103.1		
(5)			-7.8	73.8	6.5	2.0	-12.4	75.2	-16.2	76.4	-19.8	77.5	-24.5	78.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	23.6	25.7	35.1	50.1	61.0	67.3	71.1	69.7	59.4	47.2	36.3	27.4		
	DBStd	18.94	10.30	10.48	8.18	7.78	7.86	6.43	5.51	6.55	7.49	8.28	8.64	9.82		
	HDD50	3341	818	679	462	92	10	0	0	0	11	152	417	700		
	HDD65	6791	1283	1099	926	449	172	50	10	26	196	553	862	1165		
	CDD50	2595	0	0	2	95	351	518	653	611	295	65	5	0		
	CDD65	569	0	0	0	2	48	118	198	172	30	1	0	0		
	CDH74	4908	0	0	0	20	431	943	1664	1586	255	9	0	0		
	CDH80	1544	0	0	0	2	96	246	555	593	51	1	0	0		
	WSAvg	6.4	7.1	7.4	7.7	6.8	6.0	6.0	5.3	5.6	5.6	5.9	6.7	7.0		
	PrecAvg	22.3	1.7	1.3	1.5	1.6	2.0	2.3	2.6	1.7	1.9	1.9	1.9	1.8		
(15) Precipitation	PrecMax	34.0	5.4	3.1	3.8	3.9	4.3	8.0	8.7	4.1	6.5	7.8	4.7	5.0		
	PrecMin	13.3	0.2	0.0	0.2	0.4	0.5	0.2	0.6	0.2	0.1	0.0	0.2	0.1		
	PrecStd	5.0	1.1	0.7	0.8	0.9	1.1	1.5	1.8	1.0	1.4	1.7	1.1	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.9	47.3	60.7	77.1	87.1	89.2	93.3	96.4	85.3	73.5	59.6	49.2		
		MCWB	42.4	43.3	48.1	57.4	65.4	67.6	69.9	68.4	63.3	59.5	53.5	47.8		
	2%	DB	40.3	43.0	54.7	70.7	82.6	85.7	89.4	90.6	80.7	68.1	54.5	44.9		
		MCWB	38.9	40.8	45.2	54.4	62.5	67.5	69.2	67.7	61.4	57.4	50.9	43.6		
	5%	DB	37.7	39.7	50.3	67.0	79.0	82.8	86.1	86.4	76.5	62.8	51.0	42.0		
		MCWB	36.3	37.8	43.1	52.7	62.0	66.8	68.2	66.8	60.0	55.3	48.5	41.0		
	10%	DB	35.4	37.1	46.4	63.0	75.0	79.7	82.8	82.4	72.2	58.7	47.4	39.0		
		MCWB	34.3	35.5	41.0	50.9	60.3	65.3	67.4	65.4	59.0	53.2	45.4	38.0		
	0.4%	WB	42.8	44.5	49.8	59.3	68.4	72.0	73.1	72.7	66.5	62.2	54.6	48.1		
		MCDB	43.6	46.3	58.3	73.7	83.1	82.5	86.5	85.9	78.3	70.0	58.0	49.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.3	41.1	46.6	56.1	65.5	70.0	71.4	70.2	63.9	58.7	51.6	43.8		
		MCDB	40.1	42.9	52.6	66.9	77.6	81.7	84.4	84.1	75.2	65.6	54.3	44.7		
	5%	WB	36.5	38.0	43.8	54.2	63.4	68.3	70.0	68.3	62.2	56.1	48.7	41.0		
		MCDB	37.5	39.3	48.9	63.1	74.4	79.2	82.4	81.9	72.6	61.4	50.5	41.9		
	10%	WB	34.4	35.8	41.6	52.5	61.6	66.6	68.6	66.7	60.3	53.8	45.7	38.2		
		MCDB	35.3	37.0	45.9	60.7	72.1	77.1	79.9	79.5	68.7	58.1	47.2	38.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	9.2	10.9	15.7	17.7	16.5	17.4	18.6	16.3	12.8	8.2	7.4		
		MCDBR	6.8	10.1	17.6	21.0	22.7	20.5	21.7	23.5	22.8	19.1	12.1	8.3		
		MCWBR	6.4	8.4	10.8	10.1	9.1	8.8	8.0	8.0	9.6	10.1	9.0	7.7		
		MCDBR	6.7	9.3	15.6	17.7	19.8	18.2	18.9	20.3	19.0	15.9	10.8	8.2		
	5% WB	MCWBR	6.5	8.3	10.4	9.5	9.2	8.7	8.1	8.5	9.7	10.1	9.0	7.8		
(40) Clear-Sky Solar Irradiance	taub		0.318	0.347	0.378	0.446	0.410	0.401	0.431	0.429	0.401	0.365	0.348	0.321		
	taud		2.183	2.129	2.184	2.086	2.271	2.316	2.239	2.233	2.279	2.356	2.301	2.262		
	Ebn at Noon		221	244	259	253	269	272	261	256	250	239	211	204		
	Edh at Noon		25	34	38	47	41	39	42	40	35	27	23	20		
(44) All-Sky Solar Radiation	RadAvg		297	571	954	1380	1825	1959	1908	1699	1157	668	285	193		
	RadStd		57	97	134	128	155	172	120	126	147	101	54	44		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.37	N/A	N/A	+1.64	N/A	N/A	-283	-373	+198	+122
(47)	Regional (9 neighbors)	+1.39	N/A	N/A	+1.87	N/A	N/A	-265	-360	+220	+140

Nomenclature: See separate page